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System, apparatus, and method for multi-dimensional adaptive microphone-loudspeaker array sets for room correction and equalization

Abstract

In at least one embodiment, an audio system is provided. The audio system includes a plurality of loudspeaker, a plurality of microphones, and an audio controller. The plurality of loudspeakers transmits an audio signal in a listening environment. The plurality of microphones detects the audio signal in the listening environment. The at least one audio controller is configured to determine a first psychoacoustic perceived loudness (PPL) of the audio signal as the audio signal is played back through a first loudspeaker of the plurality of loudspeakers and to determine a second PPL of the audio signal as the audio signal is sensed by a first microphone of the plurality of microphones. The at least one audio controller is further configured to map the first loudspeaker of the plurality of loudspeakers to the first microphone of the plurality of microphones based at least on the first PPL and the second PPL.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application is the U.S. national phase of PCT Application No. PCT/US2020/033802 filed on May 20, 2020, the disclosure of which is incorporated in its entirety by reference herein.

TECHNICAL FIELD

(2) Aspects disclosed herein may generally relate to a system, apparatus, and method for multi-dimensional adaptive microphone-loudspeaker array sets for room correction and room equalization. In one aspect, the disclosed system, apparatus and/or method may map listening rooms into microphone and loudspeaker array sets according to criteria based on human perception of sound and psychoacoustics. These aspects and others will be discussed in more detail below.

BACKGROUND

(3) When sound is reproduced by one or more loudspeakers, the perception of the desired auditory illusion is modified by the listening environment. The sound reproduction system can also introduce undesired artifacts. Room response equalization (RRE) aims at improving the sound reproduction in rooms by applying advanced digital signal processing techniques to design an equalizer on the basis of one or more measurements of the room response. Various established techniques can be used for solving the RRE problem including, homomorphic filtering, linear predictive coding (LPC), least-squares optimization, frequency-domain deconvolution, and multiple-input/multiple-output inverse theorem (MINT) solutions. Various pre-processing methods are usually employed for improving RRE techniques such as non-uniform frequency resolution, complex smoothing, frequency warping, Kautz filters and multi-rate filters. Current room correction and equalization techniques may be categorized as single position or multiple position monitoring with fixed or adaptive room equalizers. Their complexities may increase exponentially as more features are supported. Thus, such systems may become an obstacle for successful implementation on real-time processors. Additionally, these solutions may not utilize psychoacoustics which involves the study of sound perception and audiology based on the manner in which humans perceive various sounds.

(4) Current loudspeaker-microphone array sets mapping techniques rely on proximity analysis for determining the influential loudspeakers on a given listening position. In other words, the loudspeakers are mapped to listening areas based on their physical distance from the microphones within the listening area. These techniques become inefficient in small enclosures, like car cabins, where a large number of loudspeakers may become equally close to more than one listening position, thus increasing the computational complexities and reducing the benefits of room equalization. Moreover, proximity analysis may exclude influential speakers, which are beyond a given distance from the listening position.

SUMMARY

(5) In at least one embodiment, an audio system is provided. The audio system includes a plurality of loudspeaker, a plurality of microphones, and an audio controller. The plurality of loudspeakers transmit an audio signal in a listening environment. The plurality of microphones detect the audio signal in the listening environment. The at least one audio controller is configured to determine a

first psychoacoustic perceived loudness (PPL) of the audio signal as the audio signal is played back through a first loudspeaker of the plurality of loudspeakers and to determine a second PPL of the audio signal as the audio signal is sensed by a first microphone of the plurality of microphones. The at least one audio controller is further configured to map the first loudspeaker of the plurality of loudspeakers to the first microphone of the plurality of microphones based at least on the first PPL and the second PPL.

(6) In at least one embodiment, an audio system is provided. The audio system includes a plurality of loudspeakers, a plurality of microphones, and at least one audio controller. The plurality of loudspeakers is configured to transmit an audio signal in listening environment. Each of the microphones is positioned at a respective listening location in the listening environment. The plurality of microphones is configured to detect the audio signal in the listening environment. The at least one audio controller is configured to determine a first psychoacoustic perceived loudness (PPL) for each loudspeaker of the plurality of loudspeakers and to determine a second PPL for each microphone of the plurality of microphones to employ an adaptive process for equalizing the audio signal in the listening environment.

(7) In at least one embodiment, a method for employing an adaptive process for equalizing an audio signal in the listening environment is provided. The method includes transmitting, via a plurality of loudspeakers, an audio signal in listening environment and detecting, via a plurality of microphones positioned in a listening environment, the audio signal in the listening environment. The method includes determining a first psychoacoustic perceived loudness (PPL) for each loudspeaker of the plurality of loudspeakers and determining a second PPL for each microphone of the plurality of microphones to employ an adaptive process for equalizing the audio signal in the listening environment based on the first PPL and the second PPL

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The embodiments of the present disclosure are pointed out with particularity in the appended claims. However, other features of the various embodiments will become more apparent and will be best understood by referring to the following detailed description in conjunction with the accompany drawings in which:

(2) FIG. 1 illustrates a system for providing audio for a two-dimensional arbitrary, microphone and loudspeaker room array;

(3) FIG. 2 illustrates a system for providing audio for a two-dimensional microphone and loudspeaker room array in accordance to one embodiment;

(4) FIG. 3 illustrates a method for performing a calibration to map one or more loudspeakers to one or more microphones in accordance to one embodiment;

(5) FIG. 4 illustrates a system for assigning loudspeakers to microphones during the calibration method of FIG. 3 in accordance to one embodiment;

(6) FIG. 5 illustrates a system for performing an adaptive run-time process for room correction and equalization in accordance to one embodiment; and

(7) FIG. 6 illustrates a method for performing the adaptive run-time process for the room correction and equalization system of FIG. 5 in accordance to one embodiment.

DETAILED DESCRIPTION

(8) As required, detailed embodiments of the present invention are disclosed herein; however, it is to be understood that the disclosed embodiments are merely exemplary of the invention that may be embodied in various and alternative forms. The figures are not necessarily to scale; some features may be exaggerated or minimized to show details of particular components. Therefore, specific structural and functional details disclosed herein are not to be interpreted as limiting, but

merely as a representative basis for teaching one skilled in the art to variously employ the present invention.

(9) It is recognized that the controllers as disclosed herein may include various microprocessors, microcontrollers, digital signal processors (DSPs), integrated circuits, memory devices (e.g., FLASH, random access memory (RAM), read only memory (ROM), electrically programmable read only memory (EPROM), electrically erasable programmable read only memory (EEPROM), or other suitable variants thereof), and software which co-act with one another to perform operation(s) disclosed herein. In addition, such controllers as disclosed utilizes one or more microprocessors to execute a computer-program that is embodied in a non-transitory computer readable medium that is programmed to perform any number of the functions as disclosed. Further, the controller(s) as provided herein includes a housing and the various number of microprocessors, integrated circuits, and memory devices ((e.g., FLASH, random access memory (RAM), read only memory (ROM), electrically programmable read only memory (EPROM), electrically erasable programmable read only memory (EEPROM)) positioned within the housing. The controller(s) as disclosed also include hardware-based inputs and outputs for receiving and transmitting data, respectively from and to other hardware-based devices as discussed herein.

(10) Room equalization, or correction, may be necessary for a successful immersive and high-fidelity listening experience inside enclosed spaces such as, for example, vehicle cabins. The process of room equalization (RE) involves, among other things, compensating for unwanted room sound artifacts, such as early reflections, reverb reflections, surrounding material properties, and loudspeaker imperfections. Moreover, RE may be performed in a fixed manner or in an adaptive manner. In the fixed RE, calibration is performed initially, and filter coefficients are calculated and used with minimal to no updates after calibration. In adaptive RE, calibration is performed initially to determine some initial conditions and henceforth, run-time adaptation is performed to update filter coefficients to track changing room conditions in real time.

(11) Most room enclosures are considered weakly stationary environments. For example, room conditions change as functions of room geometry (i.e., furniture, fixtures, luggage, etc.), room capacity (i.e., number of people, pets, etc.), and room environment (i.e., temperature, humidity, etc.). Therefore, it may be necessary to adapt RE filter coefficients to these changing conditions for improved performance.

(12) Aspects disclosed herein map listening rooms into microphone and loudspeaker array sets based on the human perception of sound (e.g., psychoacoustics). Following this initial calibration phase, a run-time process continuously updates equalization filter coefficients to adapt for the changing conditions for a room. The disclosed techniques involve human perception properties and are flexible enough to allow these modes of operation depending on the application. In particular, the disclosed techniques may map any number of loudspeakers to any number of listening positions. Moreover, the equalization filter coefficients may be fixed where the calibration process is performed, and the filter coefficients may be adaptive when the run-time process is performed in addition to calibration.

(13) FIG. 1 illustrates an example audio system **100**. The system **100** includes an array of loudspeakers **102a-102g** (e.g., “**102**”) positioned in a listening environment **104**. The listening environment **104** may correspond to, for example, a vehicle cabin, living room, concert hall, etc. While FIG. 1 depicts that the loudspeakers **102** surround an array of microphones, such microphones correspond to simulated listening positions of users **106a-106p** (e.g., “**106**”) in the listening environment **104**. An audio controller **108** is operably coupled to the array of loudspeakers **102** for providing an audio input via the loudspeakers **102** into the listening environment **104**. It is recognized that the locations of the loudspeakers **102** and the listening positions of users **106** may be fixed or variable. The loudspeakers **102** and the listening positions of users **106** generally form a two-dimensional array. It may be desirable to map a corresponding loudspeaker **102** to one or more listening positions **106** to enable a user to experience optimal

sound perception of the audio.

(14) FIG. 2 illustrates an audio system **200** in accordance to one embodiment. The audio system **200** includes an array of loudspeakers **202a-202f** (“**202**”) and an array of microphones **204a-204c** (“**204**”) positioned in a listening environment **205**. Each of the microphones **204a-204d** are positioned at corresponding listening positions of users **206a-206d**, respectively, in the listening environment. At least one audio controller **208** (hereafter “audio controller”) is operably coupled to the array of loudspeakers **202** for providing an audio input via the loudspeakers **202** into the listening environment **205**. The audio input includes signals with acoustic frequencies in the audible and/or ultrasonic ranges. For example, the audio input may include test signals such as sine waves, chirp waves, gaussian noise, pink noise, etc. or audio recordings. It is recognized that the locations of the loudspeakers **202** and the listening positions of users **206** may be fixed or variable. It is desirable to map a corresponding loudspeaker **202** to one or more of the listening positions **206**. The microphones **204** are illustrated and provided in the listening environment **205** to enable the audio controller **208** to perform calibration for mapping each loudspeaker **202** to one or more microphones **204** (i.e., or one or more listening positions **206**).

(15) As noted above, it is desirable to map a corresponding loudspeaker **202** to one or more of the listening positions **206** to enable the user to experience the most optimal listening experience. The mapping of a particular loudspeaker **202** to one or more of the listening positions **206** to achieve optimal audio playback (i.e., audio perception) may be based on, for example, the psychoacoustic perceived loudness (PPL) and on the distance of the loudspeaker **202** to the listening position **206**. PPL is a measure of perceptually relevant information contained in any audio record. PPL represents a theoretical limit on how much acoustic energy is perceived by the human ear at various time intervals or frames. PPL is defined as follows:

$$PPL = \sum_{k=1}^{CB} [E(k) - T(k)]$$

(16) Where $E(k)$ is the energy in the k th psychoacoustic critical band and is complex-valued, $T(k)$ is the masking threshold in the k th psychoacoustic critical band and is real-valued, and CB is the number of psychoacoustic critical bands. The masking threshold for critical band, k provides a power level under which any acoustic energy is not audible to the listener. Acoustic energy in critical band, k above the masking threshold is audible to the listener. Calculations of $E(k)$ and $T(k)$ follow techniques developed in the areas of perceptual coding of digital audio. For example, the audio signal is first windowed and transformed to the frequency domain. A mapping from frequency domain to psychoacoustic critical band domain is performed. Masking thresholds are then obtained using perceptual rules. The frequency-domain transformation is performed by first multiplying a section of the audio input, or frame, defined over a time interval, with a window function, for example Hamming, Hann, Blackman etc., followed by a time to frequency transform, such as, FFT, DFT, DCT, Wavelet, etc. The frequency-domain signal is then multiplied by a matrix of linear or non-linear mappings from frequency-domain to psychoacoustic critical band domain. The psychoacoustic critical band domain includes perceptual scales such as the equivalent rectangular bandwidth (ERB) scale, the Bark scale, or Mel scale. The masking thresholds $T(k)$ may be estimated by first calculating the power in each critical band, i.e.,

$P(k) = \text{Real}(E(k)).\text{sup.2} + \text{Imaginary}(E(k)).\text{sup.2}$, applying a spreading function (SF), and then calculating various psychoacoustic measures such as spectral flatness measure (SFM), coefficient of tonality, and masking offsets.

(17) PPL ideal (PPL_I) (or the first psychoacoustic perceived loudness) is generally calculated from the audio inputs (and at the loudspeakers **202a-202c**) at some time intervals or over the whole audio sequence. PPL measured (PPL_M) (or the second psychoacoustic perceived loudness) is calculated at the microphone inputs at similar time intervals. PPL loss (PPL_L) is the difference between PPL_I and PPL_M and measures the amount of acoustic energy deviations from ideal due to room conditions and speaker imperfections. PPL_L is calculated as complex-valued and hence contains information on both level deviations (magnitude) and time of arrival deviations (phase),

$$PPL_{sub.L} = PPL_{sub.I} - PPL_{sub.M} = \sum_{sub.k=1}^{sup.CB} [E_{sub.I}(k) - T_{sub.I}(k)]$$

$$- \sum_{sub.k=1}^{sup.CB} [E_{sub.M}(k) - T_{sub.M}(k)]$$

(18) It's reasonable to assume that measured masking thresholds, $TM(k)$, should be equal to ideal masking thresholds, $TI(k)$, so PPL_L in equation above is approximated as:

$$PPL_{sub.L} = PPL_{sub.I} - PPL_{sub.M} = \sum_{sub.k=1}^{sup.CB} [E_{sub.I}(k) - E_{sub.M}(k)]$$

(19) This approximation may avoid the computationally intensive masking threshold computations while still providing accurate results for magnitude and phase of the acoustic deviations from ideal.

(20) Thus, in order to take into account, the PPL of the loudspeaker **202** to the listening position **206** (e.g., the microphone **204**), the audio controller **208** performs the following calibration process. For each microphone **204** or “m”, the audio controller **208** measures a PPL_I from each loudspeaker **202**, while a calibration audio signal is played back in the listening environment. This results in (m*s) PPL_M quantities at the microphones, which measure the influences of room conditions and loudspeaker design on every microphone **204** according to equations:

$$|PPL_{sub.I}| = \sum_{sub.k=1}^{sup.CB} |E_{sub.I}(k) - T_{sub.I}(k)|; \text{ for } |E_{sub.I}(k)| > T_{sub.I}(k)$$

$$|PPL_{sub.M}| = \sum_{sub.k=1}^{sup.CB} |E_{sub.M}(k) - T_{sub.M}(k)|; \text{ for } |E_{sub.M}(k)| > T_{sub.M}(k)$$

$$|PPL_{sub.L}| = |PPL_{sub.I}| - |PPL_{sub.M}|$$

(21) Where $||$ is the complex magnitude operator and only energies above their respective critical band hearing thresholds are included.

(22) For each microphone **204** positioned in the array as illustrated in FIG. 2, the audio controller **208** determines the loudest set of loudspeakers within the array of loudspeakers **202** using PPL. For example, $|PPL_I|$ of the input audio waveform is calculated by the audio controller **208** over either the entire track length of the audio or at some time intervals (e.g., example every 10 milliseconds) as the audio is played sequentially through loudspeakers **202a**, **202b**, **202c**, **202d**, **202e**, and **202f**. Simultaneously, as audio is playing through each loudspeaker **202**, $|PPL_M|$ is measured at microphone **204a** over similar time intervals. For each loudspeaker **202**, $|PPL_L|$ is calculated as the difference between $|PPL_I|$ and $|PPL_M|$, which determines perceived audible deviations at the listening position **206a**. The magnitude quantity of PPL_L determines perceived audio loudness levels deviations from ideal at the listening position **206a**. A programmable threshold level of perceived loudness magnitude loss is used to discriminate between influential loudspeakers **202** at listening position **204a** from non-influential loudspeakers. The audio controller **208** may assign any given microphone **204** to one or more loudspeakers **202**. For example, the audio controller **208** may assign loudspeakers **202a** and **202b** to microphone **204a** based on the psychoacoustic perceived loudness. The audio controller **208** may assign loudspeakers **202b** and **202c** to the microphone **204b** based on loudness (e.g., based on PPL and PPL loss).

(23) FIG. 3 illustrates a method **300** for performing a calibration to map one or more loudspeakers **202** (e.g. an array of loudspeakers **202**) to one or more microphones **204** (e.g., an array of microphones **204**) in accordance to one embodiment. In operation **302**, the audio controller **208** loops over the number of microphones **204** positioned within the listening environment **205**. In this operation, the audio controller **208** stores data corresponding to the total number of microphones **204** that are positioned in the listening environment **205**.

(24) In operation **304**, the audio controller **208** loops over the number of loudspeakers **202** positioned within the listening environment **205**. In this operation, the audio controller **208** stores data corresponding to the total number of loudspeakers **202** that are positioned in the listening environment **205**.

(25) In operation **306**, the audio controller **208** calculates the $|PPL_I|$, $|PPL_M|$, and $|PPL_L|$ loss for each microphone and loudspeaker group set iteration.

(26) In operation **308**, the audio controller **208** compares $|PPL_L|$ to a programmable threshold level of perceived loudness magnitude loss which is used to discriminate between influential loudspeakers **202** at listening position **204a** from non-influential loudspeakers. As noted above, $|PPL_L|$ is calculated as the difference between $|PPL_I|$ and $|PPL_M|$, which determines perceived

audible deviations at the listening position **206a**. If the $|PPL_L|$ is less than the programmable threshold level, then the method **300** moves to operation **310**. If not, then the method **300** moves to operation **312**.

(27) In operation **310**, the audio controller **208** determines whether all of the loudspeakers **202** in the array have generated the $|PPL_I|$, $|PPL_M|$, and $|PPL_L|$ loss (e.g., operations **306** and **308** are executed for every loudspeaker **202** in the array). If this condition is met, then the method **300** moves to operation **302**. If not, then the method moves back to operation **304**.

(28) In operation **312**, the audio controller **208** assigns a corresponding loudspeaker **202** to one or more microphones **204**. In operation **314**, the audio controller **208** stores RE calibration fixed coefficients that are ascertained via from the PPL_L (i.e., the psychoacoustic perceived loudness loss). Once the loudspeaker-microphone array set mapping is complete and the fixed calibration coefficients calculated and stored, then RE is performed by applying these coefficients to the input of the loudspeakers as illustrated in FIG. **4**.

(29) FIG. **4** illustrates a system **400** for assigning loudspeakers **202** to microphones **204** in reference to operation **312** of the method **300** of FIG. **3** in accordance to one embodiment. The system **400** includes the audio controller **208** and an array **460** having the one or more loudspeakers **202** and the one or more microphones **204**. The one or more microphones **204** may be positioned proximate to corresponding listening positions of the users **206a-206b**. The audio controller **208** includes memory **209** for storing the RE fixed coefficients as derived from the method **300** during calibration. The one or more microphones **204** may be positioned proximate to corresponding listening positions of the users **206**. The audio controller **208** includes a first plurality of filter banks **450a-450b**, a matrix mixer **451**, a plurality of multiplier circuits **452a-452c**, a second plurality of filter banks **454a-454b**, and a function block **472**.

(30) With respect to the assignment of the loudspeakers **202** to the microphones **204**, the audio controller **208** may assign the loudspeakers **202a**, **202b** to the microphone **204a** at the listening position **206**. The audio controller **208** may assign the loudspeakers **202b**, **202c** to the microphone **204b** at the listening position **206b**. The first plurality of filter banks **450a-450b** may be implemented as analysis filter banks and is configured to transform a stereo audio input (e.g., inputs R and L) into the psychoacoustic critical band domain. The matrix mixer **451** generates three channels from the stereo 2-channel audio input. The calibration methodology of FIG. **3** generates 4 sets of fixed calibration coefficients ($W(M1,S1)$, $W(M1,S2)$, $W(M2,S2)$, and $W(M2,S3)$) to perform RE in the listening environment **205**. The function block **472** receives the calibration coefficients $W(M1,S2)$ and $W(M2,S3)$ and combines (or merges) the same to generate a single output which is fed to the multiplier circuit **472**. The function block **472** merges the calibration coefficients $W(M1,S2)$ and $W(M2,S3)$ to combine responses from the microphones **204a**, **204b**, such as, but not limited to, maximum, minimum, average, smoothing. The second plurality of filter banks (or synthesis filter banks) **454a-454c** are configured to filter outputs (e.g., compensated signals) from the multiplier circuits **452a-452c**, respectively. The compensated signals are transformed back into the time-domain with the synthesis filter bank **454a-454c** before being sent to the speakers **202a-202c**.

(31) Referring back to FIG. **3**, in operation **314**, the audio controller **208** stores the assignments of the one or more loudspeakers **202** to each microphone **204** in memory **209** thereof. As noted above in connection with FIG. **2**, each of the microphones **204a-204d** are positioned at corresponding listening positions (or locations) of users **206a-206d**, respectively, in the listening environment **205**.

(32) FIG. **5** illustrates a system **500** for performing an adaptive run-time process for room correction and equalization to occur in real time in accordance to one embodiment. In contrast, the calibration process as disclosed in connection with FIG. **3** is static in terms of mapping the one or more loudspeakers **202** to one or more microphones **204** (or listening positions **206**) to enable a user to experience optimal sound perception of the audio. It is recognized that numerous room conditions change as functions of room geometry (i.e., furniture, fixtures, luggage, etc.), room

capacity (i.e., number of people, pets, etc.), and room environment (i.e., temperature, humidity, etc.) dynamically impact the listening experience for users in the listening environment **205**. The system **500** is generally configured to employ a continuous run time algorithm to account for these changing room conditions. This may be advantageous for, but not limited to, a listening environment within a vehicle.

(33) The system **500** generally includes various features as set forth in the system **400** of FIG. **4** (e.g., the audio controller **208**, the first plurality of filter banks **450a-450b**, the matrix mixer **451**, the plurality of multiplier circuits **452a-452c**, a second plurality of filter banks **454a-454b**, and the array **460**. The system **500** further includes a plurality of first delay blocks **453a-453c**, a plurality of first delay blocks **456a-456c**, a first plurality of psychoacoustic modeling blocks **458a-458c**, a first plurality of difference blocks **459a-459c**, a third plurality of filter banks **461a-461b**, a second plurality of psychoacoustic modeling blocks **462a-462b**, a plurality of comparators **470a-470d**, and a function block **482**.

(34) The adaptive process performed by the system **500** may start with each of the microphones **204a-204b** providing an audio input signal to the plurality of filter banks **450a-450b**, respectively. In this case, the microphones **204a-204b** generate outputs indicative of the audio being played in the listening environment **205** via playback from the loudspeakers **202a-202c**. For example, and as noted in FIG. **4**, the loudspeakers **202a** and **202b** may be assigned to the microphone **204a** (or to listening position **206a**) and the loudspeakers **202b** and **202c** may be assigned to the microphone **204b** (or to listening position **206b**). As noted above, the plurality of filter banks (or analysis filter bank) **450a-450b** transforms the stereo audio input into audio in a psychoacoustic critical band domain. The matrix mixer **451** generates three channels from the stereo 2-channel audio input. The plurality of first delay blocks **453a-453c** delay the outputs from the matrix mixer **451**.

(35) The compensation circuits **452a-452c** are generally configured to compensate either the magnitude or phase (or both the magnitude and phase) of the audio input received. The second plurality of filter banks **454a-454c** are configured to filter outputs from the compensation circuits **452a-452c**, respectively. The plurality of filter banks **454a-454c** are configured to transfer the compensated signals from the compensation circuits **452a-452c** into the time-domain prior to the audio being transmitted to the loudspeakers **202a-202c**. The loudspeakers **202a-202c** playback the audio as provided by the second filter blocks **254a-254c** into the listening environment **205**. The microphones **204a-204b** sense the audio as played back in the listening environment **205** and outputs the sensed audio to the third plurality of filter banks (or analysis filter banks) **461a-461b**, respectively for filtering. The psychoacoustic modeling blocks **462a-462b** convert the filtered, sensed audio and calculate an energy in each critical sub-band of a psychoacoustic frequency band which is represented by $EM(m,j)$, where m corresponds to the number of microphones and j corresponds to the critical band in the psychoacoustic frequency scale from critical band number 1 to critical band number CB covering an audible acoustic frequency range, for example from 0 to 20 kHz. For example, the psychoacoustic modeling blocks **462a-462b** generates $EM(1, j)$ and $EM(2, j)$, respectively. The psychoacoustic modeling block **462a** provides $EM(1, j)$ to the comparators **470a, 470b**. The psychoacoustic modeling block **462b** provides $EM(2, j)$ to the comparators **470c** and **470d**. The relevance of the comparators **470a-470d** will be discussed in more detail below.

(36) While the audio input is provided to the first plurality of filter banks **450a-450b**, the audio input is also provided to the delay blocks **456a-456b**. The delay blocks **456a-456b** delay the audio input by, for example, 10 to 20 msec. The delayed audio input is provided to the psychoacoustic modeling blocks **458a-458c**. The delay blocks **456a-456c** are applied to both microphone and loudspeaker paths to provide frame synchronization between both the microphone and loudspeaker paths. It is recognized that tuning of delay values that are utilized in the delay blocks **456a-456c** may be necessary to achieve frame synchronization between both the microphone and loudspeaker paths (e.g., there will be a delay between when the loudspeaker **202** plays back the audio and then the microphone **204** captures the audio that is played back via the loudspeaker **202**). The

psychoacoustic modeling blocks **458a-458c** convert the delayed audio inputs and calculate an energy in each critical sub-band of a psychoacoustic frequency band which is represented by $ES(s,j)$, where s corresponds to the number of loudspeakers and j corresponds to the critical band number in the psychoacoustic frequency scale from critical band number 1 to critical band number CB covering the audible acoustic frequency range, for example from 0 to 20 kHz.

(37) For the loudspeaker **202a**, the comparator **470a** generates sub-band coefficient $WS(s,j)$ or $WS(1,j)$ which generally corresponds to a difference between the psychoacoustic frequency band for the loudspeaker **202a** and the microphone **204a**. For example, the sub-band coefficient $WS(1,j)=ES(1,j)-EM(1,j)$ which is transmitted to the compensation circuit **452b** to modify the audio input to the loudspeaker **202a**. Similarly for loudspeaker **202b**, the comparator **470b** generates sub-band coefficient $WS.sub.1(s,j)$ or $WS.sub.1(2,j)$ which generally corresponds to a first difference between the psychoacoustic frequency band for the loudspeaker **202b** and the microphone **204a**. For example, the sub-band coefficient $WS.sub.1(2,j)=ES(2,j)-EM(1,j)$ which is transmitted to the function block **482**. Additionally, for the loudspeaker **202b**, the comparator **470c** generates sub-band coefficient $WS.sub.2(s,j)$ or $WS.sub.2(2,j)$ which generally corresponds to a second difference between the psychoacoustic frequency band for the loudspeaker **202b** and the microphone **204b**. The sub-band coefficient $WS.sub.2(2,j)=ES(2,j)-EM(2,j)$ is transmitted to the function block **482**. The function block **482** combines responses for both microphones **204a**, **204b** by either taking a maximum, minimum, average, or smoothing, etc. of the responses for both microphones **204a**, **204b**. The function block **482** transmits an output which corresponds to the function of the psychoacoustic frequency band for the loudspeaker **202b** and for both microphones **204a**, **204b** to the compensation circuit **452c** to modify the audio input to the loudspeaker **202b**.

(38) For loudspeaker **202c**, the comparator **470d** generates sub-band coefficient $WS(s,j)$ or $WS(3,j)$ which generally corresponds to a difference between the psychoacoustic frequency band for the loudspeaker **202c** and the microphone **204b**. For example, the sub-band coefficient $WS(3,j)=ES(3,j)-EM(2,j)$ which is transmitted to the compensation circuit **452a** to modify the audio input to the loudspeaker **202c**. The compensation circuits **452a**, **452b**, **452c** applies a complex factor (e.g., via phase or magnitude).

(39) The above adaptive process provides room equalization, or correction, which may provide for a successful immersive and high-fidelity listening experience inside enclosed spaces such as, for example, vehicle cabins. The process of room equalization (RE) involves, among other things, compensating for unwanted room sound artifacts, such as early reflections, reverb reflections, surrounding material properties, and loudspeaker imperfections.

(40) Psychoacoustic perceived loudness (or PPL), which is the subjective perception of sound pressure, can be calculated using different techniques such as equal loudness contours, absolute threshold of hearing (ATH), A-weighting, K-weighting relative to full scale (LKFS), etc. PPL may also be calculated using the psychoacoustic definitions presented herein. The advantage of embodiments disclosed herein is the ability to obtain both magnitude and phase information for the room impairments through the complex nature of the critical sub-band analysis. For example, the loudspeaker **202a** has the following transmitted and received loudness at the listening position **206a** of the microphone **204a**, respectively:

$$PPL.sub.TX=\sum.sub.j\Delta S(1,j)=\sum.sub.j|ES(1,j)-TS(1,j)| \quad (Eq. 1)$$

$$PPL.sub.RX=\sum.sub.j\Delta M(1,j)=\sum.sub.j|EM(1,j)-TS(1,j)| \quad (Eq. 2)$$

(41) Eq. 1 may be executed by the difference blocks **459a-459c** as noted in connection with FIG. 5 above. The PPL as defined in equations 1 and 2 (e.g., for the adaptive process) are similar to those referenced above in connection with the static calibration. For example, PPL.sub.TX is similar to PPL ideal (PPL_I) (or the first psychoacoustic perceived loudness) and PPL.sub.RX is similar to PPL_M (or the second psychoacoustic perceived loudness) (e.g., each of PPL ideal (PPL_I) and PPL_M have been noted above). The PPL as referenced in connection with equations 1 and 2 are being redefined for purposes of brevity. $ES(s,j)$ generally corresponds to the critical sub-band in the

psychoacoustic frequency range for the loudspeakers and TS(s,j) generally corresponds to the psychoacoustic hearing threshold for each critical sub-band. If $\Delta S(s,j)$ is greater than 0, then the audio content in the sub-band j is audible to the listeners. If $\Delta S(s,j)$ is less than 0, then the audio content in the sub-band j is not audible to the listeners.

(42) PPL.sub.Loss due to room sound artifacts is defined as:

$$PPL.sub.Loss = PPL.sub.TX - PPL.sub.RX = ES(1,j) - EM(1,j) \quad (\text{Eq. 3})$$

(43) Eq. 3 may be determined or executed by the various comparators **470a-470d** as noted in connection with FIG. 4. PPL.sub.Loss is similar to PPL_L as noted above (i.e., the psychoacoustic perceived loudness loss).

(44) PPL.sub.Loss is a complex quantity with information on both magnitude and phase as exhibited directly below.

$$Mag(PPL.sub.Loss) = |ES(1,j) - EM(1,j)| \quad (\text{Eq. 4})$$

$$(45) \text{ Phase}(PPL_{Loss}) = \arctan\left(\frac{\text{Imaginary}(ES(1,j) - EM(1,j))}{\text{Real}(ES(1,j) - EM(1,j))}\right) \quad (\text{Eq. 5})$$

(46) If PPL.sub.Loss at critical sub-band j has positive magnitude, then the dominant room impairments are due to absorption and/or dissipation. On the other hand, if PPL.sub.Loss at critical sub-band j has negative magnitude, then the dominant room impairments are due to reflections and reverberation.

(47) For the case when PPL.sub.Loss at critical sub-band j is positive in magnitude, then room equalization might involve amplifying the attenuated critical sub-bands. In general, the compensator circuits **452a-452c** may determine whether the PPL.sub.Loss at the critical sub-band j has a positive magnitude which then amplifies the audio input that is transmitted to the loudspeakers **202a-202c**, respectively.

(48) For the case when PPL.sub.Loss at critical sub-band j is negative in magnitude, room equalization might involve attenuating the amplified critical sub-bands. In general, the compensator circuits **452a-452c** may determine whether the PPL.sub.Loss at the critical sub-band j has a negative magnitude which then amplifies the audio input that is transmitted to the loudspeakers **202a-202c**, respectively.

(49) For the case when the phase of PPL.sub.RX at critical sub-band j is different than the phase of PPL.sub.TX at critical sub-band j by over a certain threshold (or predetermined threshold), then phase correction can be applied by rotating the received critical sub-band phases to match their transmitted counterparts. This complex multiplication is performed as noted above. For example, the compensation circuits **454a-454c** perform the complex multiplication when the phase of PPL.sub.RX at critical sub-band j is different than the phase of PPL.sub.TX at critical sub-band j by over a certain threshold (or predetermined threshold). If a particular loudspeaker **202** is shared by more than one microphone **204**, then a mathematical operation is performed on PPL.sub.RX microphone phases, such as maximum, minimum, average, smoothing, etc. An example of this is the first function block **472** as illustrated in FIG. 4 as the loudspeaker **202b** is shared by the microphones **204a** and **204b**.

(50) FIG. 6 illustrates a method **600** for performing the adaptive run-time process for the room correction and equalization system **500** of FIG. 5 in accordance to one embodiment. In operation **602**, the audio controller **208** determines the PPL for each loudspeaker **202a**, **202b**, **202c** in the array. For example, the audio controller **208** determines the PPL for each loudspeaker **202a**, **202b**, **202c** based on Eq. 1 as noted above. The audio controller **208** also determines the PPL for each microphone **204a**, **204b** in the array based on Eq. 2 as noted above.

(51) In operation **604**, the audio controller **208** determines the PPL loss due to sound artifacts that may be present in the listening environment **205**. For example, the audio controller **208** determines the PPL loss attributed to sound artifacts based on Eq. 3 as noted above. In operation **606**, the audio controller **208** determines whether a magnitude of the PPL loss for the loudspeakers **202a-202c** and the microphones **204a-204b** is positive or negative. For example, in operation **606**, the audio

controller **208** determines the magnitude of the PPL loss based on Eq. 4 and then determines whether such magnitude is positive or negative. If the magnitude is positive, then the method **600** proceeds to operation **610**. If the magnitude is negative, then the method **600** proceeds to operation **612**.

(52) In operation **610**, the audio controller **208** determines that the PPL loss at the critical sub-band j has a positive magnitude and that dominant listening room impairments are due to absorption and/or dissipation that is present in the listening environment **205**. In this case, the audio controller **208** amplifies the audio input provided to the loudspeakers **202a-202c** in the listening environment **205**.

(53) In operation **612**, the audio controller **208** determines that the PPL loss at the critical sub-band j has a negative magnitude and that the dominant listening room impairments are due to reflections and reverberation in the listening environment **205**. In this case, the audio controller **208** attenuates the audio input provided to the loudspeakers **202a-202c** in the listening environment **205**.

(54) In operation **614**, the audio controller **208** determines the phase of the PPL loss for the microphones **204a-204b** and the loudspeakers **202a-202c** based on eq. 5 as noted above. For example, the audio controller **208** determines whether the phase of the PPL loss for the loudspeakers **202a-202c** is different than the phase of the PPL loss for the microphones **204a-204b** by a predetermined threshold. If this condition is true, then the method **600** moves to operation **618**. If not, then the method **600** moves to back to operation **602**. In operation **618**, the audio controller **208** applies a phase correction to either the critical sub-band phases of the loudspeakers **202a-202c** (e.g., $ES(s,j)$) or the critical sub-band phases of the microphones **204a-204b** by rotating the received critical sub-band phases (e.g., the critical sub-band phases of the microphones **204a-204b**) to match their transmitted counterparts (e.g., the critical sub-band phases of the loudspeakers **202a-202c**).

(55) While exemplary embodiments are described above, it is not intended that these embodiments describe all possible forms of the invention. Rather, the words used in the specification are words of description rather than limitation, and it is understood that various changes may be made without departing from the spirit and scope of the invention. Additionally, the features of various implementing embodiments may be combined to form further embodiments of the invention.

Claims

1. An audio system comprising: a plurality of loudspeakers configured to transmit an audio signal in a listening environment; a plurality of microphones, each being positioned at a respective listening location in the listening environment, the plurality of microphones is configured to detect the audio signal in the listening environment; at least one audio controller being configured to: determine a first psychoacoustic perceived loudness (PPL) of the audio signal as the audio signal is played back through a first loudspeaker of the plurality of loudspeakers; determine a second PPL of the audio signal as the audio signal is sensed by a first microphone of the plurality of microphones; and map the first loudspeaker of the plurality of loudspeakers to the first microphone of the plurality of microphones based at least on the first PPL and the second PPL.
2. The audio system of claim 1, wherein the at least one audio controller is further configured to determine a first magnitude of the first PPL and a second magnitude of the second PPL prior to mapping the first loudspeaker to the first microphone.
3. The audio system of claim 2, wherein the at least one audio controller is further configured to obtain a difference between the first magnitude of the first PPL and the second magnitude of the second PPL prior to mapping the first loudspeaker to the first microphone.
4. The audio system of claim 3, wherein the difference between the first magnitude of the first PPL and the second magnitude of the second PPL corresponds to a PPL loss which is indicative of perceived audible deviations at a listening position in the listening environment.
5. The audio system of claim 4, wherein the audio controller is further configured to compare the

PPL loss to a programmable threshold to determine whether to map the first loudspeaker to the first microphone.

6. The audio system of claim 5, wherein the audio controller is further configured to map the first loudspeaker to the first microphone in response to the PPL loss being less than the programmable threshold.

7. The audio system of claim 1, wherein the audio controller is further configured to apply an adaptive process to equalize the audio signal in the listening environment based at least one the first PPL and the second PPL.

8. The audio system of claim 7, wherein the audio controller is further configured to determine the first PPL of the audio signal as the audio signal is played back through each loudspeaker of the plurality of loudspeakers and to determine the second PPL for each microphone of the plurality of microphones of the audio signal as the audio signal is sensed by each of the microphones of the plurality of microphones.

9. The audio system of claim 8, wherein the audio controller is further configured to determine a PPL loss for each loudspeaker of the plurality of loudspeakers and for each microphone of the plurality of microphones based on a difference between the first PPL and the second PPL.

10. The system of claim 9, wherein the at least one audio controller is further configured to amplify an audio input signal to the plurality of loudspeakers to account for absorption and/or dissipation that is present in the listening environment in the event a magnitude for the PPL loss for each of the loudspeakers and each of the microphone is positive.

11. The system of claim 9, wherein the at least one audio controller is further configured to attenuate an audio input signal to the plurality of loudspeakers to account for reflections and reverberation in the listening environment in the event a magnitude for the PPL loss for each of the loudspeakers and each of the microphones is negative.

12. The system of claim 9, wherein the at least one audio controller is further configured to determine whether a phase for the PPL loss for each of the loudspeakers is different from the PPL for each of the microphones by a predetermined amount.

13. The system of claim 12, wherein the at least one audio controller is further configured to apply a phase correction to critical sub-band phases of the plurality of loudspeakers or to critical sub-band phases of the plurality of microphones in the event the phase for the PPL loss for each of the loudspeakers is different from the PPL for each of the microphones by the predetermined amount.
